

2.9 Quantifiers in Predicate Logic

Quantifiers are used in predicate logic to describe how many objects satisfy a condition.

Why we use Quantifiers?

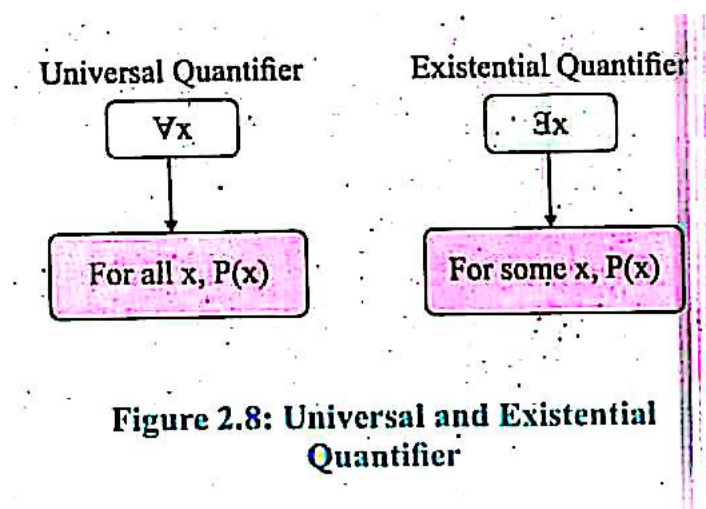
- They help us to make general statements or statements about specific objects like “All students are hardworking” or “Some books are interesting.”
- They make predicate logic more powerful than propositional logic.
- They help represent knowledge clearly and precisely.

Universal Quantifier (for all)

- Symbol: $\forall$
- Meaning: The statement is true for **all** objects in a given set.
- Example: $\forall x, x > 0 \rightarrow$ “For all x, x is greater than 0”
- It is used to make general statements about every object in a domain.

Existential Quantifier (there exists)

- Symbol: $\exists$
- Meaning: The statement is true for **at least one** object in a set.
- Example: $\exists x, x \text{ is even} \rightarrow$ “There exists at least one x that is even”
- It is used when a condition is satisfied by some object but not necessarily all.



Using Quantifiers in Logical Statements

Quantifiers can be combined with predicates to make clear logical statements.

- $\forall x, P(x) \rightarrow$ Predicate $P(x)$ is true for all x
- $\exists x, P(x) \rightarrow$ Predicate $P(x)$ is true for at least one x
- Combined example: $\forall x \exists y, x < y \rightarrow$ For every x there exists a y that is greater than x

Quantifiers allow us to express rules, conditions, and relationships clearly.

Short Questions

1. What are quantifiers in predicate logic?
2. What is the universal quantifier? Write its symbol.
3. What is the existential quantifier? Write its symbol.
4. Give one example of a statement using the universal quantifier.
5. Give one example of a statement using the existential quantifier.
6. What is the difference between universal and existential quantifiers?
7. Why are quantifiers important in predicate logic?
8. Write the meaning of $\forall x, P(x)$ and $\exists x, P(x)$.